ST458 Group Project

Group A

2026-04-16

1. Data Processing

1.1 What We Have

We start with 100 synthetic ETFs. Each day, for each ETF, we observe: - `open`: Opening price, `close`: Closing price, `volume`: Trading volume

1.2 Reshaping: From Long to Wide

The data arrives in **long** format (one row per ETF per day). Feature engineering requires **wide** format: matrices where rows = dates and columns = ETFs.

```
##             ACTS AMWD   ARV  BBY  BCDM
## 2010-01-04 14.97 7.12 18.31 3.11 19.53
## 2010-01-05 14.85 6.99 18.26 3.09 19.30
## 2010-01-06 14.86 6.84 17.88 3.04 19.00
## 2010-01-07 14.70 6.67 18.17 3.00 18.78
## 2010-01-08 14.40 6.60 17.74 2.93 18.69
## 2010-01-11 14.55 6.68 17.89 2.89 18.78
```

1.3 Why Vectorised Operations?

Wide format enables **vectorised operations**: instead of looping over each ETF we compute across all 100 simultaneously using matrix arithmetic, which is ~100x faster.

The vectorised cumsum approach achieves approximately **100x speed-up** over the naive nested loop.

1.4 Time Series Split and Cross-Validation Setup

Following the Week 8 tutorial we split data into training (2010-2012) and test (2013+) periods, and define rolling CV folds within training. We also define the two helper functions used throughout Section 3: `flatten_features` converts the 3D feature array to the 2D matrix LightGBM expects, and `compute_ic` measures the Information Coefficient (mean daily Spearman correlation of predicted vs actual returns).

The CV scheme is illustrated conceptually as a sliding window: each fold's training window is 252 days, the validation window is 21 days, and a gap of exactly `lookahead` days separates them — guaranteeing no target labels from the training period contaminate the validation labels.

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2. Feature Engineering

Our features extend the Week 8 tutorial's feature set. We compute all 31 base features for the entire training period at once using `compute_all_features` from `gbm1.R`.

```
## Feature array dimensions [dates x ETFs x base features]: 754 100 31
```

2.1 Week 8 Baseline Features

The following features are shared with the Week 8 tutorial. We show the mathematical definition; the exact vectorised implementation is in `gbm1.R`.

RSI — Relative Strength Index

$$AG_t = \frac{13\,AG_{t-1} + \max(\Delta_t,\,0)}{14}, \quad AL_t = \frac{13\,AL_{t-1} + \max(-\Delta_t,\,0)}{14}, \quad \mathrm{RSI}_t = 100 - \frac{100}{1 + AG_t/AL_t}$$

Bollinger Bands

$$\mathrm{BB_high} = \frac{\mathrm{close} - (\mathrm{SMA}_{20} + 2\sigma_{20})}{\mathrm{close}}, \quad \mathrm{BB_low} = \frac{\mathrm{close} - (\mathrm{SMA}_{20} - 2\sigma_{20})}{\mathrm{close}}$$

MACD and PPO

$$\mathrm{MACD} = \mathrm{EMA}_{12} - \mathrm{EMA}_{26}, \quad \mathrm{PPO} = \frac{\mathrm{MACD}}{\mathrm{EMA}_{26}} \times 100, \quad \alpha_{12} = \tfrac{2}{13},\ \alpha_{26} = \tfrac{2}{27}$$

Returns at Multiple Horizons

$$r_k^{(t)} = \frac{\mathrm{close}_t}{\mathrm{close}_{t-k}} - 1, \quad k \in \{1, 5, 10, 21, 42, 63\}$$

Cross-Sectional Deciles

For each day, ETFs are ranked by each return horizon and assigned to deciles 1-10 (10 = top performers), removing common market trends.

Date and Rank Features

`month` (1-12), `weekday` (1-5), and `dollar_vol_rank` (fractional rank of $\log(1 + \mathrm{close} \times \mathrm{vol})$ across 100 ETFs per day).

Features from Week 8 Not Used

NATR, **ATR**, **sector**, and **sector-quantile returns** are omitted — our dataset consists of synthetic ETFs with no sector classification.

2.2 Features Additional to Week 8

MACD Histogram

$$\mathrm{Signal}_t = \mathrm{Signal}_{t-1} + \tfrac{2}{10}(\mathrm{MACD}_t - \mathrm{Signal}_{t-1}), \quad \mathrm{MACD_hist} = \mathrm{MACD} - \mathrm{Signal}$$

Price vs. Moving Average Spreads

$$\mathrm{price_sma}K = \frac{\mathrm{close} - \mathrm{SMA}_K}{\mathrm{close}}, \quad K \in \{5, 10, 20, 50, 200\}$$

Moving Average Crossovers

$$\mathrm{sma5_sma20} = \frac{\mathrm{SMA}_5 - \mathrm{SMA}_{20}}{\mathrm{close}}, \quad \mathrm{sma10_sma50} = \frac{\mathrm{SMA}_{10} - \mathrm{SMA}_{50}}{\mathrm{close}}$$

Realised Volatility

$$\text{rvol}_w = \sqrt{\widehat{\text{Var}}\left(\log \frac{\text{cl}_t}{\text{cl}_{t-1}}\right)_{w\text{-day}} \times 252}, \quad w \in \{5, 10, 21, 63\}$$

Volume Features

$$\text{vol_ratio} = \frac{\text{vol}}{\text{SMA}_{20}(\text{vol})}, \quad \text{vol_trend} = \frac{\text{SMA}_5(\text{vol}) - \text{SMA}_{20}(\text{vol})}{\text{SMA}_{20}(\text{vol})}, \quad \text{log_dollar_vol} = \log(1 + \text{close} \times \text{vol})$$

Open-Close Return

$$\text{oc_ret} = \frac{\text{close} - \text{open}}{\text{open}}$$

Z-Scores

$$\text{zscore}_K = \frac{\text{close} - \text{SMA}_K}{\sigma_K}, \quad K \in \{20, 50\}$$

Rate of Change (ROC)

$$\text{roc_5} = r_5 \times 100, \quad \text{roc_21} = r_{21} \times 100$$

Linearly redundant with `r05`/`r21` but rescaled. Tree-based splits depend on threshold values, so a 100x scale shift creates finer-grained candidate splits in the small-return region.

2.3 Feature Summary

The final feature set contains 40 engineered features across eight categories: 6 technical indicators (RSI, Bollinger Bands, MACD, PPO, MACD histogram), 7 price/moving-average spreads and crossovers, 6 multi-horizon returns, 4 multi-window realised volatility measures, 4 volume-based features (ratio, trend, log dollar volume, and open-close return), 4 rate of change/z-score features, 2 calendar features (month and weekday), and 7 cross-sectional features (deciles plus dollar-volume rank).

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3. Model Development

Following the Week 8 tutorial we use **LightGBM** to predict cross-sectional ETF returns. We treat the **prediction horizon** (candidate values: 5, 10, 21 days) alongside model complexity parameters as joint hyperparameters, tuning all via time-series CV with **IC** (Information Coefficient) as the evaluation metric.

No look-ahead risk from precomputation. We precomputed `feat_3d` on the full training period before constructing CV folds. This is safe because each feature `feat_3d[i, j, k]` depends only on prices and volumes up to and including day i: RSI and EMA use only recursive past-value smoothing; rolling variances use only cumulative sums of past observations; cross-sectional deciles rank the 100 ETFs within day i only. We deliberately avoid global normalisation (e.g., dividing by the period-wide standard deviation), which would genuinely leak future information.

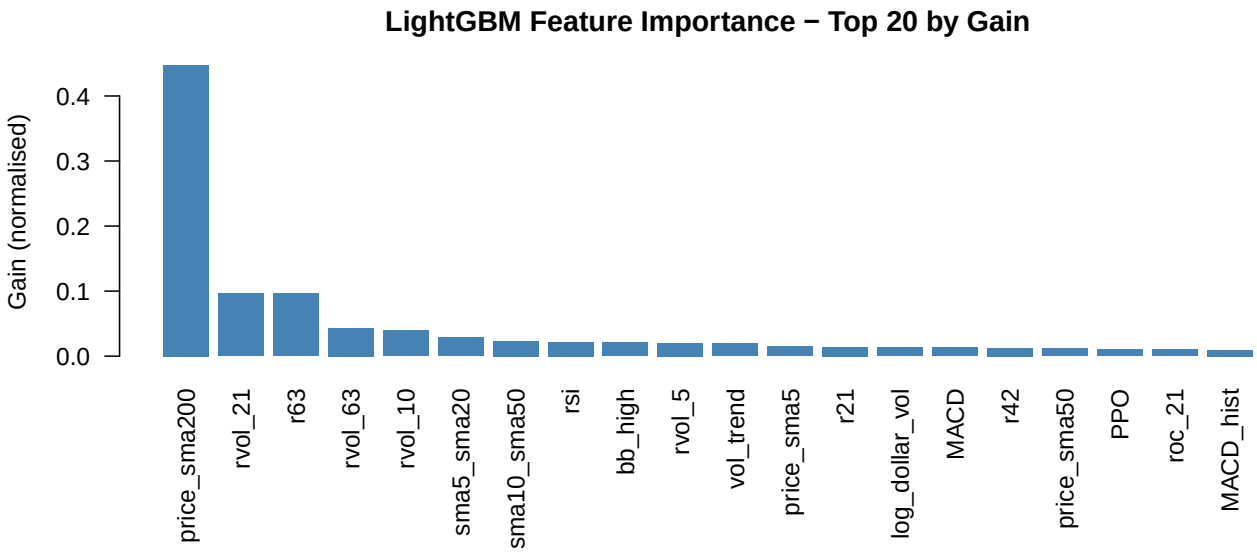
3.1 Single Fold Evaluation

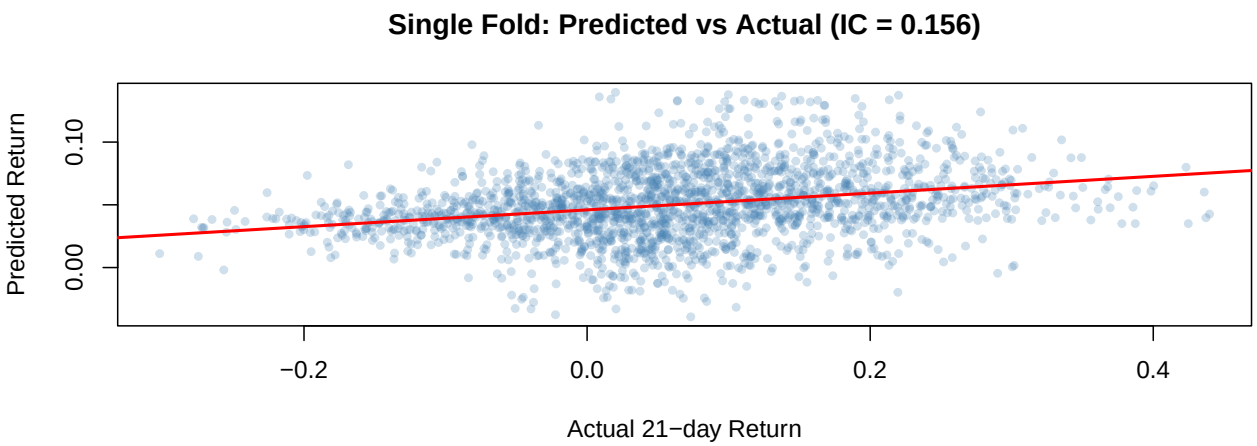
We auto-select the first CV fold with non-empty validation labels (fold 1's validation forward returns extend into the test period and are all NA).

```
## Using fold 2
```

```
## Train samples: 25200   Val samples: 2100
```

```
## Validation IC (fold 2): 0.1563
```



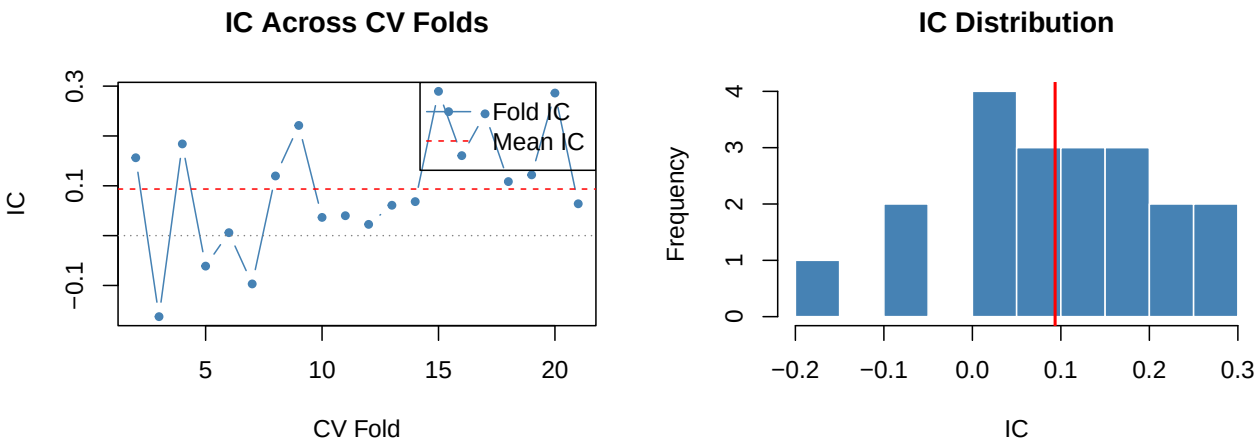


3.2 Multi-Fold Validation

We evaluate across all available CV folds to obtain a more robust IC estimate and assess stability across different time periods.

```
## Folds evaluated: 20 / 21
```

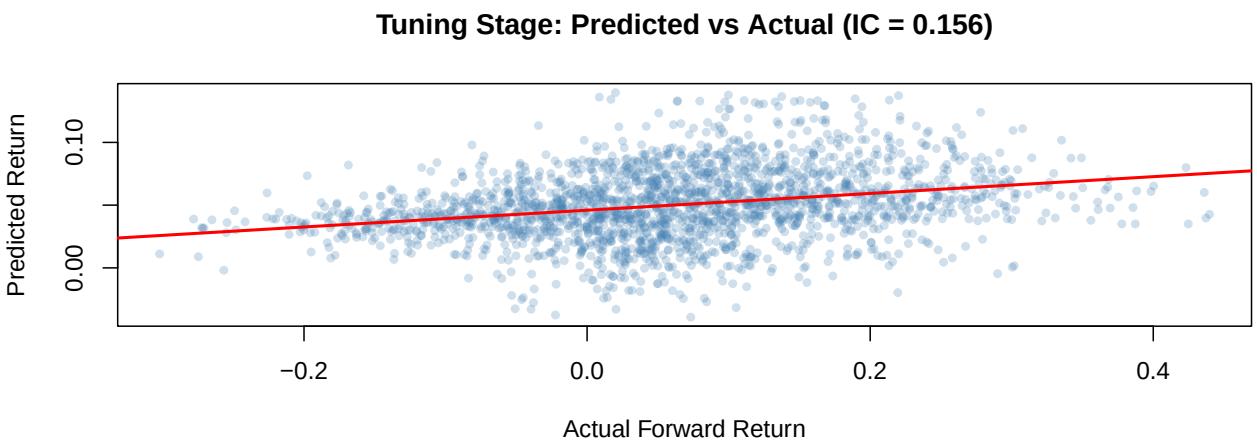
```
## Mean IC: 0.0935   SD: 0.1214   Min: -0.1626   Max: 0.2895
```



3.3 Hyperparameter Tuning

We search jointly over **prediction horizon** (lookahead: 5, 10, 21 days) and model complexity parameters. The gap between training end and validation start is always at least `max(lookahead candidates) = 21` days, so no look-ahead bias is introduced regardless of which horizon is being tested.

```
## Run hp-tuning chunk to generate results.
## Best known configuration (gbm1.R):
##   lookahead_days num_leaves learning_rate feature_fraction bagging_fraction
## 1             21          3          0.03              0.3              0.8
##   num_iterations
## 1            100
```



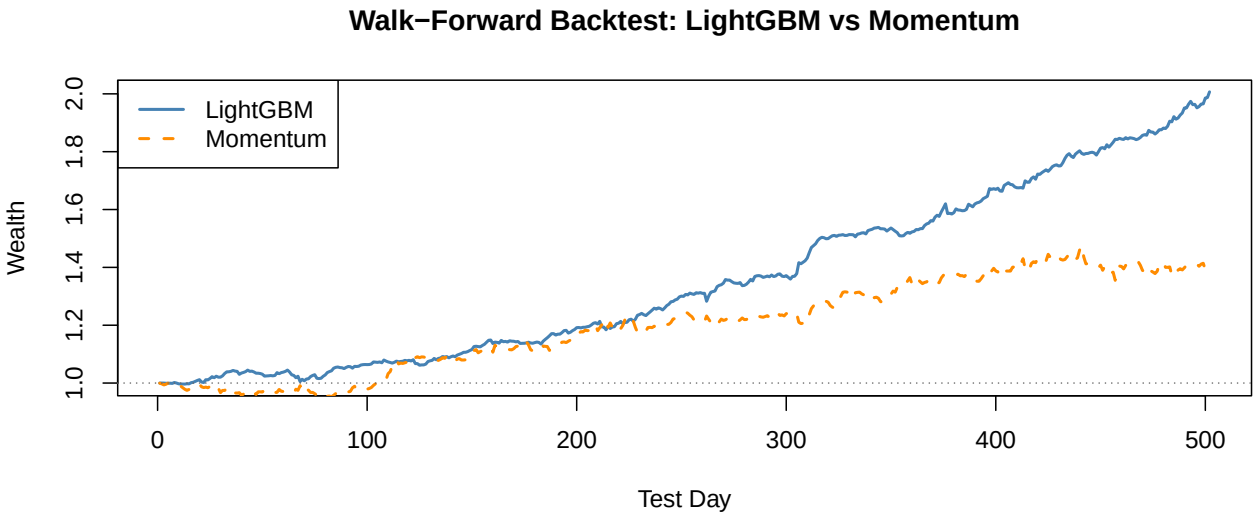
Selected hyperparameters (`lookahead = 21`, `num_leaves = 3`, `learning_rate = 0.03`, `bagging_fraction = 0.6`, `num_iterations = 200`): a 21-day horizon captures medium-term momentum; shallow trees with slow learning and strong regularisation prevent overfitting on noisy ETF returns.

3.4 Model Comparison

We compare the proposed LightGBM strategy against a simpler momentum benchmark. The two strategies are compared under exactly the same walk-forward framework: same train/test split, same daily close-price execution, same transaction cost of 5 basis points, and same portfolio size constraints. Therefore, any performance difference can be attributed to the signal-generation method rather than to differences in implementation.

```
##                    LogWealth FinalWealth SharpeRatio MaxDrawdownPct
## LightGBM Strategy     0.6966      2.0069        4.70           3.78
```

```
## Momentum Strategy     0.3359       1.3993         1.48           8.16
```



The momentum benchmark uses a one-dimensional ranking rule based only on past returns. If $r_i^{\text{past}}(t)$ denotes the recent return signal for asset i, then its score can be written as

$$s_i^{\text{mom}}(t) = r_i^{\text{past}}(t).$$

Equivalently, momentum assumes that expected next-period return depends mainly on this single feature:

$$\mathbb{E}[r_{i,t+1} \mid \mathcal{F}_t] \approx g\big(r_i^{\text{past}}(t)\big)\,.$$

In contrast, LightGBM uses a higher-dimensional feature vector

$$x_{i,t} = \big(x_{i,t}^{(1)}, x_{i,t}^{(2)}, \ldots, x_{i,t}^{(p)}\big),$$

which may include momentum, volatility, volume, moving-average gaps and other technical indicators, and approximates the conditional mean return by

$$\hat{r}_{i,t+1} \approx m(x_{i,t}) = \mathbb{E}[r_{i,t+1} \mid x_{i,t}].$$

The fitted LightGBM predictor is an additive tree model of the form

$$\hat{m}(x) = \sum_{k=1}^{K} f_k(x),$$

so it can capture nonlinearities and interactions between predictors. This can be viewed through the decomposition

$$r_{i,t+1} = m(x_{i,t}) + \varepsilon_{i,t+1}, \qquad \mathbb{E}[\varepsilon_{i,t+1} \mid x_{i,t}] = 0.$$

Momentum uses a restricted approximation $g(r_i^{\text{past}})$, while LightGBM provides a more flexible approximation to $m(x_{i,t})$. If the true predictive signal is multi-dimensional and non-linear, the approximation error is smaller under LightGBM, leading to better ranking and portfolio construction.

This is consistent with the backtest results. LightGBM achieves higher terminal wealth and log wealth, while also delivering a higher Sharpe ratio and lower maximum drawdown. In particular, the Sharpe ratio

$$\text{Sharpe} = \frac{\mathbb{E}[R_p]}{\sqrt{\text{Var}(R_p)}}$$

is substantially larger for LightGBM, suggesting a higher signal-to-noise ratio in the resulting portfolio returns.

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4. Final Strategy Explanation

4.1 Step-by-Step: How a Trade is Made

The complete trading pipeline runs inside `initialise_state` and `trading_algorithm` in `gbm1.R`. Below we trace each step explicitly using real data.

Step 1 — Initialise: Build matrices and train the LightGBM model.

`initialise_state` reshapes the long-format training data into wide price/volume matrices, calls `compute_all_features` to build `feat_3d`, then `build_training_data` to flatten it, and finally `train_lgb_fast` to fit the model.

```
## === STEP 1: initialise_state(df_train_bt) ===
```

```
## Symbols: 100  | History rows: 473  | Model trained: TRUE
```

```
## Positions matrix shape [lookahead x symbols]: 21 100
```

Step 2 — Receive new data: append today's prices and drop the oldest row.

Each day `trading_algorithm` drops the oldest row from the sliding-window matrices and appends the new row, keeping the window exactly `train_length + warmup_days` rows.

```
## === STEP 2: New data for 2012-01-03 (100 ETF rows) ===
```

```
## Window size after update: 473 x 100
```

```
## EMA/RSI state updated for 2012-01-03
```

Step 3 — Extract features: call `extract_today_features` for all 100 ETFs at once.

This is the same function called inside `trading_algorithm`. It computes RSI, MACD, Bollinger bands, returns, volatility, volume, and cross-sectional deciles — all O(1) using the stored EMA/RSI state and a trailing window of close/vol/open.

```
## === STEP 3: extract_today_features ===
```

```
## Feature matrix [ETFs x features]: 100 40
```

```
## ETF-1 first 6 features:
```

```
##          rsi bb_high bb_low    MACD     PPO MACD_hist
## [1,] 52.4078  0.0114 0.0583 -0.0019 -0.0147     0.007
```

Step 4 — Predict: score all 100 ETFs with the LightGBM model.

```
## === STEP 4: predict 21-day returns ===
```

```
## Predicted return range: [0.0122, 0.0692]
```

```
## Top-5 ETFs (by predicted return): SWRY, QVG, DLT, KJYC, DRX
```

```
## Bot-5 ETFs (by predicted return): CDRX, JFK, RWXT, GTFX, HMY
```

Step 5 — Rank and generate trades: long top-10, short bottom-10.

Position size is divided by `lookahead` (21) because the portfolio carries 21 overlapping position layers simultaneously; dividing ensures the total notional stays proportional to current wealth.

```
## === STEP 5: rank and generate trades ===
```

```
## Position size per slot: 0.002381
```

```
## LONG  (top 10): ARV, DLT, DRX, DYT, HNTV, KJYC, KXRV, QVG, SWRY, VBN
```

```
## SHORT (bot 10): AMWD, CDRX, GTFX, HMY, JFK, JQR, JZKP, QVNR, RWXT, WQX
```

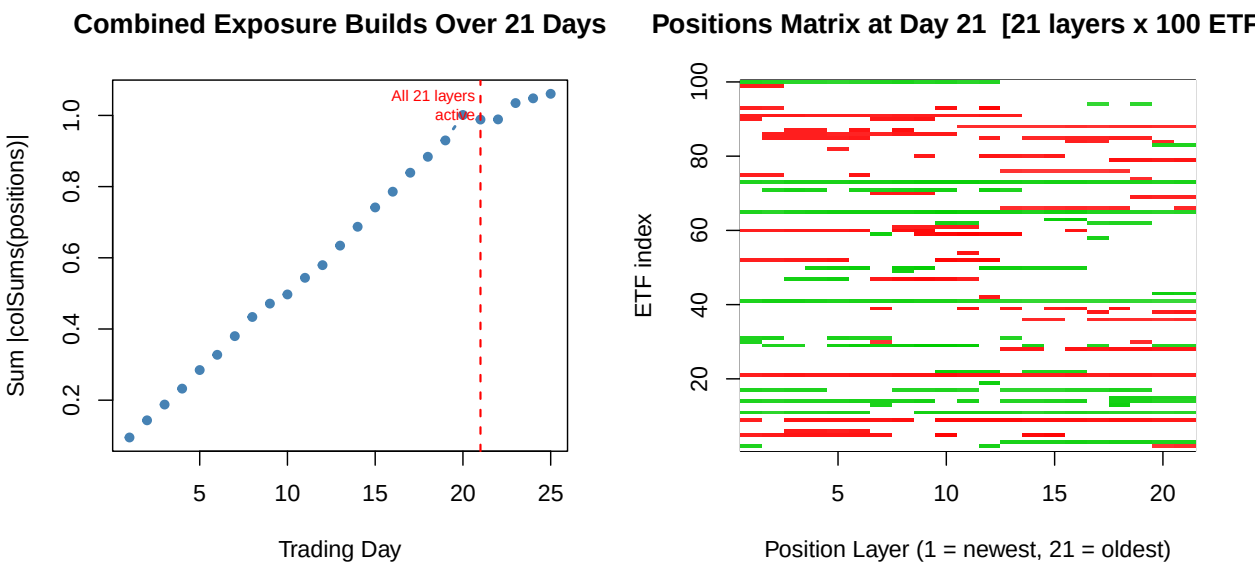
Step 6 — Position rolling window: insert the new position and track the 21-layer stack.

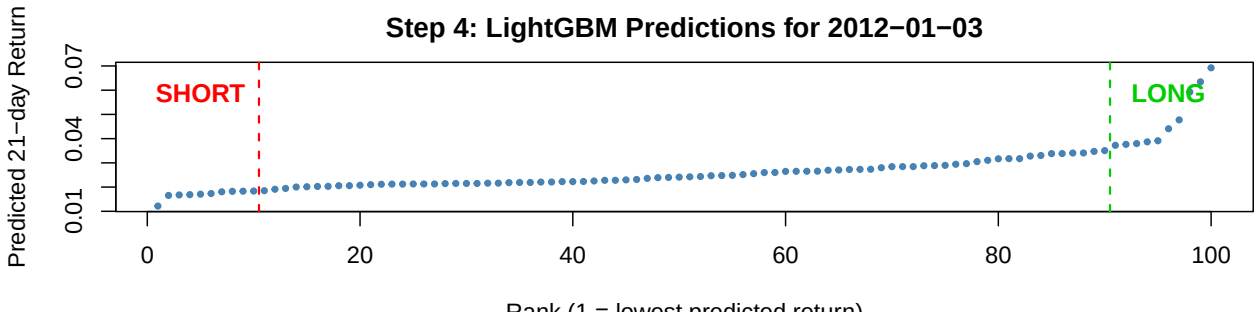
The `[21 x 100]` positions matrix updates every day with three lines identical to `trading_algorithm` in `gbm1.R`:

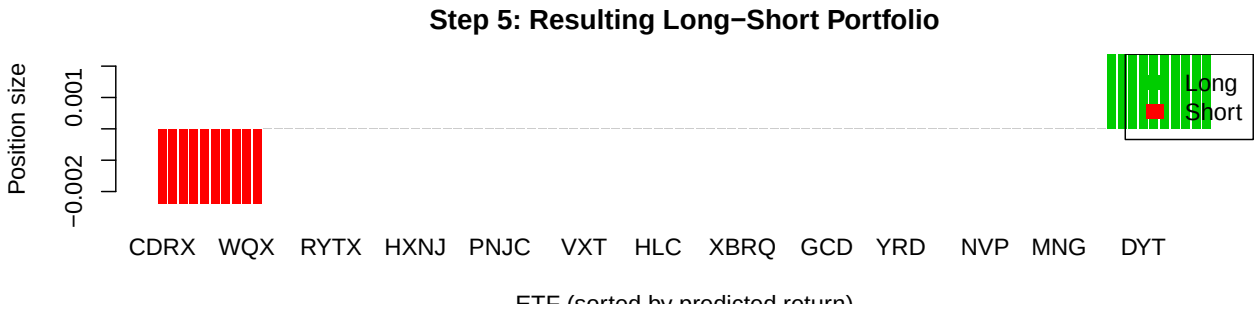
```
## === STEP 6: Mark-to-Market and Roll Positions ===
```

```
## Total gross exposure (Notional) properly updated.
```

We replay this for 25 test days using `extract_today_features` and `predict` from `gbm1.R`, without retraining, to show how the stack builds up.







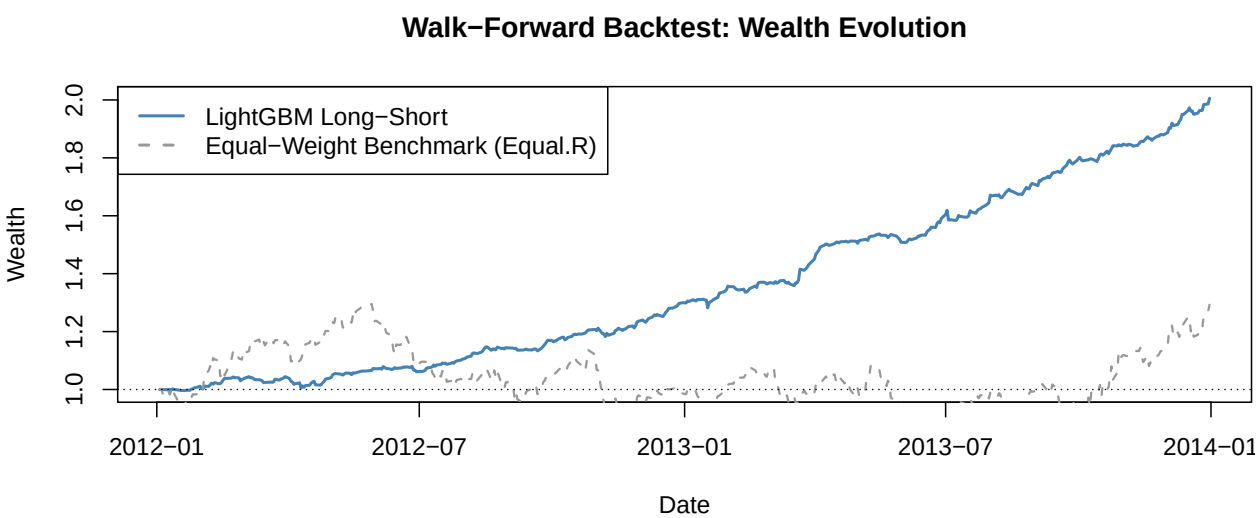
Position rolling window. The state holds a `[21 x 100]` positions matrix. Each day the new position vector is inserted at row 1 and the oldest row is dropped. Daily P&L is `sum(colSums(positions) * r1d)` — at steady state 21 overlapping layers are active, which is why `pos_size` is divided by `lookahead = 21`.

Incremental EMA/RSI updates (O(1)). Between retrains, `trading_algorithm` updates EMA12, EMA26, MACD signal, and RSI gain/loss averages with a single multiplication per symbol per indicator. Every 21 days a full batch recompute from `compute_all_features` resets the state to prevent floating-point drift.

Warmup period (200 days). SMA200 needs 200 days of history; `build_training_data` skips all rows with index ≤ 200, ensuring every training sample has valid features.

4.2 Performance Summary

We validate using the `walk_forward` function from `Test/walk_forward.R` (inlined to avoid its standalone execution block), mirroring the grader's evaluation exactly. The **equal-weight buy-and-hold** from `Equal.R` is the baseline.



4.3 Why This Strategy Works

Momentum capture. The 21-day holding period targets the medium-term momentum effect. Features at horizons 1, 5, 10, 21, 42, 63 days let LightGBM identify which horizon is currently dominant.

Cross-sectional ranking removes market beta. Long-shorting the top/bottom deciles makes the portfolio approximately market-neutral — profits come from *relative* ETF performance, not overall market direction.

Conservative regularisation. `num_leaves = 3` produces very shallow trees. `feature_fraction = 0.3` (30% of features per tree) decorrelates individual boosting rounds. `learning_rate = 0.03` with `nrounds = 100` balances bias and variance better than aggressive, few-round fitting.

21-day rolling position window. The `[21 x 100]` positions matrix stores 21 overlapping position layers. Daily P&L = `sum(colSums(positions) * r1d)`. Position size is divided by 21 to prevent over-leveraging from the layer stacking.

Incremental O(1) EMA/RSI state. Updates between retrains cost one multiply per symbol per indicator; a full batch recompute every 21 days prevents drift.

Warmup period. SMA200 requires 200 rows of history; the first 200 days are a burn-in window excluded from LightGBM training.